Friday, 19 December 2025 6:06 PM

- **Descriptive statistics**
  - It helps us to describe historical data.
  - It helps us to build reports | summary.
  - It helps us to get visual dashboards.

- **Inferential statistics.**
  - The word inferential itself means estimating or approximating something.
  - Why we are doing so?
    - Data collection is super expensive, time consuming activity.

- Quiz What is the average age of an Indian citizen.

Gender	age	Height
→		
→		
→		
→		
→		

n=5

In pandas, it is super easy.

$$\frac{age_1 + age_2 + age_3 + age_4 + age_5}{5}$$

- But to compute the average age of citizen we need the data for the entire population.

In India → 140 crores or 1.4 billion people (approximation).

So, if we have the data for all the 1.4B people

$$\frac{age_1 + age_2 + \ldots\ldots\ldots + age_{1.4B}}{1.4B}$$ → pandas

Challenges

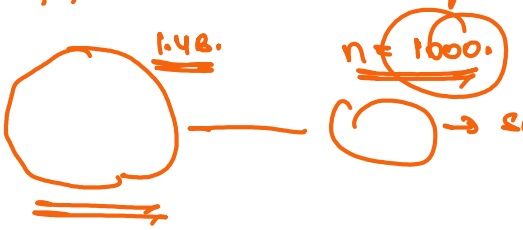
- Collect the data of 1.4 Billion people.
  - → 2-3 years (many born, many will die).
  - → expensive
  - → Risk (error).
  - → Time consuming.

Applied pandas → mean

spark → mean.

Do I have the data with me → No.

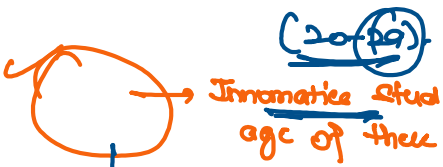
So, we are going to take samples out of the entire population so that we can able to either predict or estimate or approximate about a population based on a sample of data.



→ sample size depends
- how much your organization is willing to spent.

Try to go as big as possible to collect the data and the data that we have collected should be collected at random.

Suppose



→ Innomatics Students & compute the average age of these student

↳ is this sample the correct representation of the population → No

Collect representative sample

↳ it is a subset of the population that actually reflects the characteristics / property for the entire population approximately.

If the data is a correct representation of the Indian pop.

correct

point estimation → $\mu_{population} \approx \bar{x}_{sample}$ → Inference.

Sampling → to give each and every individual a fair chance to be considered in the sample.

↳ to avoid biasness.

- why point estimates is considered a weak way to estimate
  - No measure of uncertainity.

mean sal = 40000

↳ range

  - Highly sensitive to sample variations.

→ different results → unstable.

estimate.

- Can be misleading.

  A point estimate may be too far from the true population parameter. especially.
  - small sample size
  - skewed data
  - Presence of outliers.

- No confidence interval.
  - Point estimate does not answer → how confident are we in this estimate.

- Poor in decision making

  In real time application for medicines, MRI, stats, business, we need.

  best estimate + range.

- Sampling techniques
  - Convenience Sampling.
  - Systematic sampling.
  - Simple random sampling.
  - Stratified sampling.
  - Cluster sampling.
  - Snowball sampling.

- Population Parameters and Sample Statistics

  Population Parameters → numerical characteristics of an entire population
  - describe some aspects of the population's distribution or behaviour.

  eg. population mean, population standard deviation, population proportion etc.

  Sample Statistics → numerical measures calculated from a sample of the population
  - They are used to estimate or infer information about the corresponding population parameters.

  eg sample mean, sample variance, sample standard deviation.

- Population mean (μ) and sample mean ($\bar{x}$).

$$\text{Population mean } (\mu) = \frac{\sum_{i=1}^{N} x_i}{N}$$

Sample mean $(\bar{x}) = \frac{\sum_{i=1}^{n} x_i}{n}$

- Population variance (σ^2) and Sample variance (s^2).
  - Population variance $(\sigma^2) = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$
  - Sample variance $(s^2) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$

20 → variation

19 variation

$x = [4, 6, 8]$

$\bar{x} = \frac{4+6+8}{3} = \frac{18}{3} = 6$

first deviation $= 4-6 = -2$
second $= 6-6 = 0$
third deviation $= 8-6 = 2$

$-2+0+2 = 0$

$\frac{1}{n-1}$ → Bessel's correction factor.

One degree of freedom → already used to estimate the mean.

- Population standard deviation & sample standard deviation
  - Population standard deviation $(\sigma) = \sqrt{\sigma^2}$

$$= \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

  - Sample standard deviation $(s) = \sqrt{s^2}$

$$= \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$